Implied PDF and Implied Volatility

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Abstract

We recall that the implied volatility given by a particular case of the Fokker–Planck equation can be derived – according to Dupire's approach – from a continuum of call prices.

We assume that the market is composed of a risk-free asset whose value at time t is S_t^0, solution of $\mathrm{d}S_t^0 = rS_t^0\,\mathrm{d}t$, and a risky asset whose value at time t is S_t, solution of $\mathrm{d}S_t = rS_t\,\mathrm{d}t + \sigma(t,S_t)\,\mathrm{d}W_t$. We suppose that we have a continuum of call prices $C(T,K)$ for different maturities T and strikes K.

According to Bruno Dupire's paper *Pricing with a Smile*, we have the following result – a particular case of the Fokker–Planck equation relating $C(T,K)$ to $\sigma(T,K)$, stated here for $r=0$ (it generalises easily to $r\neq 0$).

Theorem 1.

$$\sigma^2(T,K) = \frac{2\,\dfrac{\partial C(T,K)}{\partial T}}{\dfrac{\partial^2 C(T,K)}{\partial K^2}}$$

Proof. We have $C(T,K) = \mathbb{E}^{P^*}[(S_T-K)^+] = \mathbb{E}^{P^*}\left[(S_T-K)\,\mathbf{1}_{\{S_T>K\}}\right]$, hence:

$$\frac{\partial C(T,K)}{\partial K} = -\mathbb{E}^{P^*}[\mathbf{1}_{\{S_T>K\}}] = -P^*(S_T>K) = P^*(S_T<K) - 1 = \int_{-\infty}^{K} \varphi_{S_T}(K')\,\mathrm{d}K' - 1$$

$$\frac{\partial^2 C(T,K)}{\partial K^2} = \varphi_{S_T}(K)$$

where φ_{S_T} denotes the probability density function of S_T.

Since $r=0$, $\mathrm{d}S_t = \sigma(t,S_t)\,\mathrm{d}W_t$. Using Itô's lemma, for every function f of class C^2 with $f(S_0)=0$:

$$f(S_T) = \int_0^T f'(S_t)\,\mathrm{d}S_t + \frac{1}{2}\int_0^T f''(S_t)\,\sigma(t,S_t)^2\,\mathrm{d}t$$

Taking expectations, the stochastic integral vanishes:

$$\mathbb{E}[f(S_T)] = \frac{1}{2}\int_0^T \mathbb{E}\left[f''(S_t)\,\sigma(t,S_t)^2\right]\mathrm{d}t = \frac{1}{2}\int_0^T \int_{\mathbb{R}} f''(K)\,\varphi_{S_t}(K)\,\sigma(t,K)^2\,\mathrm{d}K\,\mathrm{d}t$$

Differentiating with respect to T:

$$\frac{\partial \mathbb{E}[f(S_T)]}{\partial T} = \frac{1}{2}\int_{\mathbb{R}} f''(K)\,\varphi_{S_T}(K)\,\sigma(T,K)^2\,\mathrm{d}K = \frac{1}{2}\int_{\mathbb{R}} f''(K)\,\frac{\partial^2 C(T,K)}{\partial K^2}\,\sigma(T,K)^2\,\mathrm{d}K$$

Lemma. For every function f of class C^2 with $f(-\infty)=0$:

$$f(S_T) = \int_{\mathbb{R}} f''(K)\,(S_T-K)^+\,\mathrm{d}K$$

Proof of the lemma.

$$\int_{\mathbb{R}} f''(K)(S_T-K)^+ \,\mathrm{d}K = \int_{-\infty}^{S_T} f''(K)(S_T-K)\,\mathrm{d}K = -\int_{-\infty}^{S_T} f'(K)\,\mathrm{d}K = f(S_T)-f(-\infty) = f(S_T) \qquad \square$$

(the middle equality follows from integrating by parts once, using $(S_T - K)\big|_{K=S_T} = 0$).

Using the lemma:

$$\mathbb{E}[f(S_T)] = \int_{\mathbb{R}} f''(K)\,\mathbb{E}\big[(S_T-K)^+\big]\,\mathrm{d}K = \int_{\mathbb{R}} f''(K)\,C(T,K)\,\mathrm{d}K$$

$$\Rightarrow \quad \frac{\partial \mathbb{E}[f(S_T)]}{\partial T} = \int_{\mathbb{R}} f''(K)\,\frac{\partial C(T,K)}{\partial T}\,\mathrm{d}K$$

Equating the two expressions obtained for $\dfrac{\partial \mathbb{E}[f(S_T)]}{\partial T}$, for every admissible f:

$$\int_{\mathbb{R}} f''(K)\,\frac{\partial C(T,K)}{\partial T}\,\mathrm{d}K = \int_{\mathbb{R}} f''(K)\,\sigma^2(T,K)\,\frac{\partial^2 C(T,K)}{\partial K^2}\,\mathrm{d}K$$

Since this holds for every such f (hence for arbitrary f''), we conclude:

$$\frac{\partial C(T,K)}{\partial T} = \frac{1}{2}\sigma^2(T,K)\,\frac{\partial^2 C(T,K)}{\partial K^2} \quad \Longrightarrow \quad \sigma^2(T,K) = \frac{2\,\partial C(T,K)/\partial T}{\partial^2 C(T,K)/\partial K^2} \qquad \square$$

Bibliography

References

[1] Dupire, B. (January 1994). Pricing with a Smile. *Risk Magazine*.